

Pipelined Processor. (Note)

— In pipelined processor, instruction execution process is divided into five stages. Each stage completed in one clock cycle. so that multiple instruction can be executed simultaneously.

— Stages of pipelined processor

- ① Fetch ② Decode ③ execute ④ memory access
⑤ writeback

① Fetch (instruction fetch)

processor reads (fetch) the instruction from the memory.

② Decode (instruction decode)

the instruction is decoded, control signals are generated and read the operands from the register file

eg: add s_1, s_2, s_3

① decode the instruction and by using instruction bits (op, funct3, funct7) decode the instruction

② Read the content of Register file (content of s_2 & s_3).

③ Execution (Ex)

ALU perform the computation such as arithmetic and logical operations.

④ Memory access (memory)

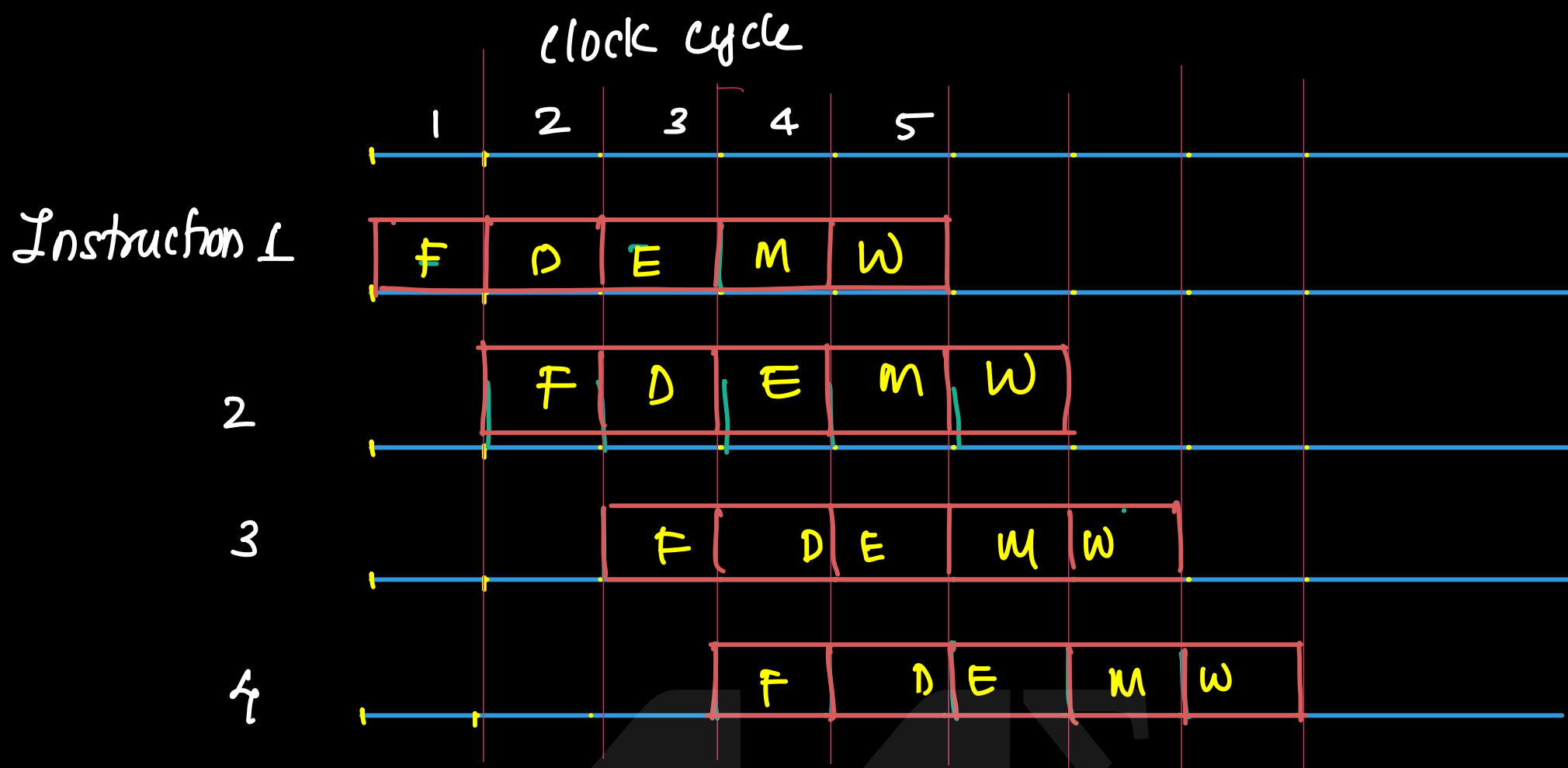
Data memory is accessed to read or write data

⑤ Write back

Final result written to the register file, if needed.

Timing diagram of pipelined process (Performance Comparison of single cycle and pipelined processor)

— Consider 4 instructions (I_1, I_2, I_3, I_4)



— In First cycle, I_1 is fetched from the memory

→ In 2nd clock cycle, I_1 enter to the decode stage and I_2 is fetched

→ In 3rd clock cycle, I_1 executes, I_2 enter to decode stage and I_3 is fetched

→ In 4th clock cycle, I_1 enter to the memory access stage, I_2 executes, I_3 enter to the decode stage, I_4 is fetched

→ In 5th clock cycle

I_1 → write back the result in Register file

I_2 → access the data memory

I_3 → execute

I_4 → Decode

— at the end of 5th clock cycle First instruction executed

→ at the end of 6th clock cycle 2nd instruction executed

→ after 5th clock cycle each instruction is completed in each clock cycle.

i.e. 1 instruction in 1 clock cycle

$$\text{clock cycle time} = 200\text{ps} = 200 \times 10^{-12} \text{ s}$$

$$1 \text{ instruction} \longrightarrow 200 \times 10^{-12} \text{ s}$$

$$\lambda \longleftarrow 1 \text{ s}$$

$$\lambda = \frac{1}{200 \times 10^{-12}} = 5000 \times 10^6 = 5 \times 10^9 = 5 \text{ billion}$$

i.e. In this pipelined processor can execute 5 billion instruction in one second.

i.e. throughput = 5 billion instruction per second.

Total time taken for the single cycle process to completing one instruction = time for Fetch + time for decode + Time for execution + Time for memory access + Time for writeback

$$T = 200\text{ps} + 100\text{ps} + 120 + 200 + 60 = 680\text{ps}$$

one instruction in 680×10^{-12} second

$$1 \text{ instruction} \longrightarrow 680 \times 10^{-12} \text{ s.}$$

$$\lambda \longleftarrow 1 \text{ s}$$

$$\lambda = \frac{1}{680 \times 10^{-12}} = 1470 \times 10^6 = 1.47 \times 10^9 = 1.47 \text{ billion}$$

For single cycle processor can complete 1.47 billion instruction in one second

throughput = 1.47 billion instruction per second
Single

through put of pipelined processor = 5 billion instruction per second.

So through put of pipelined processor = 3 times of through put of single cycle processor.



Operation of Pipelined processor (Note)

→ In pipelined processor, instruction execution process is divided in 5 stages.

→ each pipeline stage is represented with its major components

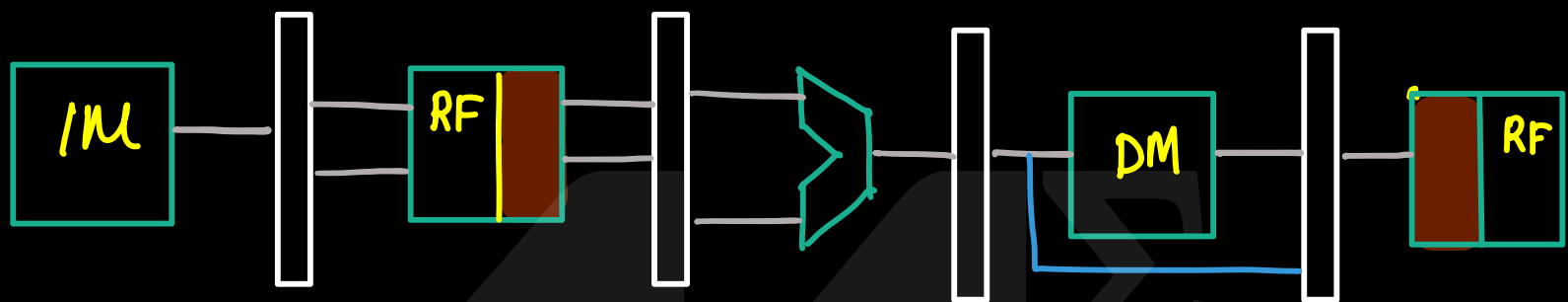
Fetch — Instruction Memory (IM)

Decode — Register file (RF)

Execution — ALU

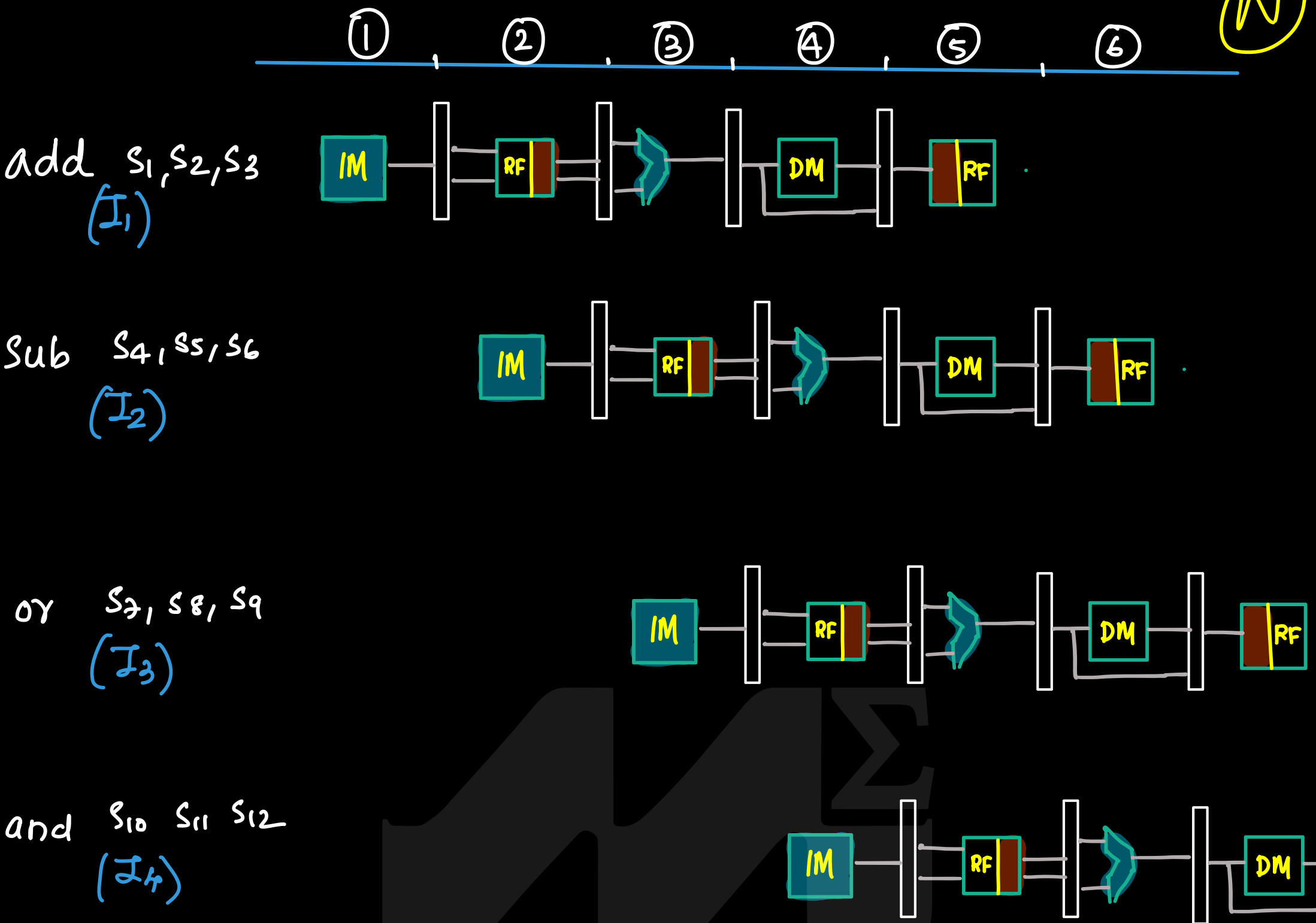
Memory — Data Memory (DM)

Write Back — Register file



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— In first clock cycle, I_1 is fetched from the IM.

→ In 2nd clock cycle
 I_1 reads s_2, s_3 in the second half cycle of cycle 2
 I_2 fetched from the memory

→ In 3rd clock cycle
 I_1 → execute and result will be available at ALU output
 I_2 — reads s_5, s_6
 I_3 — fetch the instruction

→ In 4th clock cycle
 I_1 → Since this instruction is not a load/store instruction. So Data memory is idle (not used.)

$I_2 \rightarrow$ execute subtraction

$I_3 \rightarrow$ Reads s_8, s_9

(N)

$\rightarrow$ In 5th clock cycle

$I_1 \rightarrow$ store the result in register file in the first half of cycle 5. after first half of cycle 5 result will be available in register s_1

$I_2 \rightarrow$ not accessing DM

$I_3 \rightarrow$ execute the instruction

Execution of First instruction completed at the end of cycle 5. (i.e. sum is available in Register s_1 after the end of first half of cycle 5).

